



Remittance API Design Specification Document

Version 3.0

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Contact Person

|       |                                 |
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## Change History

| Change Date | Version | Author        | Change Description                                             |
|-------------|---------|---------------|----------------------------------------------------------------|
| 2017/05/15  | 3.0     | Andrew Mlamba | Add API Endpoint,<br>Remove custom specs for<br>Western Union. |
| 2017/06/06  | 3.0     | Andrew Mlamba | Add status enquiry<br>method.                                  |

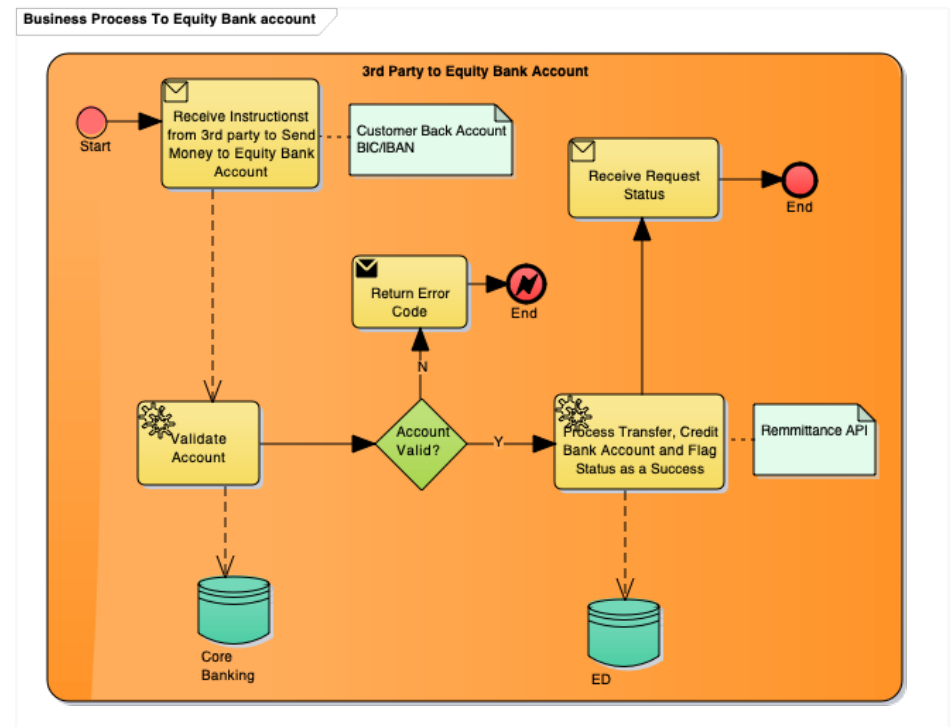
## 1.0 Introduction

Remittance API is a webservice API exposed to third party money transfers organisations to facilitate transfers of funds to:

- Equity Bank Customers Kenya
- Other Banks Customers in Kenya
- Mobile Wallets In Kenya
- Mobile Wallets in the Subsidiaries
- To Equity Bank Subsidiaries Countries(Rwanda,Uganda,South Sudan and Tanzania)

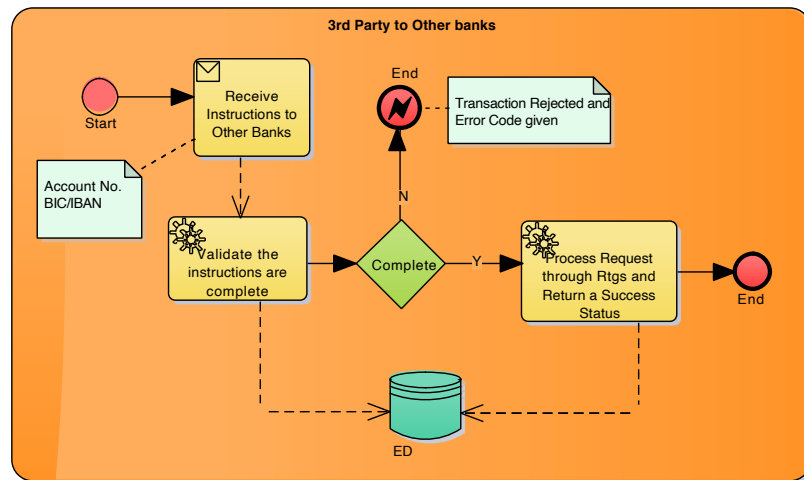
## 1.1 Transaction Process Flows

*To Equity Bank*



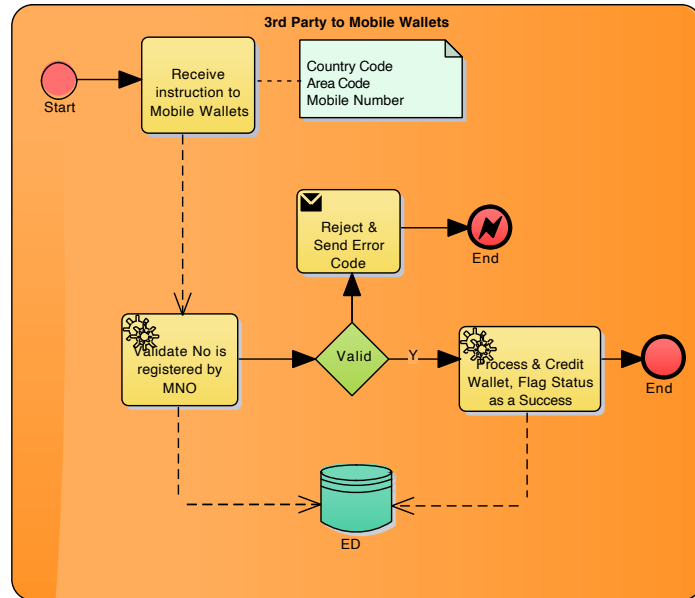
*To Other Banks*

**Business Process To Other Banks**



### To Mobile Wallets

#### Business Process To Mobile Wallets





## 1.2 API Description

Endpoint: <https://wsuat.equitybankgroup.com/v1/transactionprocessor>

The API is SOAP webservice published on HTTPS and is accessible via a secure VPN. The requests and response are XML format.

The following are the fields description:

| Service        | Operation              | Dataservices            |
|----------------|------------------------|-------------------------|
| Remittance API | TransApply             | Process Funds transfers |
| Status API     | TranApplyRequestStatus | Get transaction Status  |

### 1.2.1 Remittance API

*Request Field Mapping*



| Source Service Tag | Data Type   | Length | Variable Length | Mand /Opt | Description Default/Expected Values                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------|-------------|--------|-----------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AccountNumber      | String/text | 13     | Y               | M         | Beneficiary Bank Account Number                                                                                                                                                                                                                                                                                                                                                                                                           |
| PaymentReference1  | String/text | 100    | Y               | M         | Payment reference number e.g. Unique Transaction code/Reference. This field will be used to reconcile and batch proofing.                                                                                                                                                                                                                                                                                                                 |
| PaymentReference2  | String/text | 100    | Y               | M         | Additional Payment reference e.g. Beneficiary Name or the Account Name                                                                                                                                                                                                                                                                                                                                                                    |
| PaymentReference3  | String/text | 100    | Y               | M         | Additional Payment reference e.g. Sender details or source transaction reference number                                                                                                                                                                                                                                                                                                                                                   |
| PaymentType        | Lookup      | 50     | Y               | M         | Static - The name of the remittance agent e.g FUNTECH                                                                                                                                                                                                                                                                                                                                                                                     |
| Remarks            | String/text | 100    | Y               | O         | The detail of the recipient of the funds i.e. the name of the beneficiary or the name of the account where money will be credited.                                                                                                                                                                                                                                                                                                        |
| AgentID            | Number      | 10     | Y               | M         | A unique number to used to identify the paying account (to be shared)                                                                                                                                                                                                                                                                                                                                                                     |
| Check              | String/text | 200    | Y               | M         | This field is a checksum. The transaction acquiring organization/agent will sign this field with their private Key. First the following fields are concatenated. (SALT + (AccountNumber +BIC) or (Wallet +Mobile Number) )+ paymentReference1+ paymentType+ agentID+ tranAmount+ currencyCode+messageID) in that order (Salt to be shared) (Mandatory). The resulting text is then signed with Private Key and Base64 encoded.(Mandatory) |
| Data               | String/text | 200    |                 | O         | Not used for now(Future use)                                                                                                                                                                                                                                                                                                                                                                                                              |
| MessageID          | String/text | 100    |                 | M         | A unique number for every call to the web service                                                                                                                                                                                                                                                                                                                                                                                         |
| Fees               | Float       | -      |                 | M         | Not used for now(Future use) default to Zero                                                                                                                                                                                                                                                                                                                                                                                              |





|              |             |     |   |   |                                                                                 |
|--------------|-------------|-----|---|---|---------------------------------------------------------------------------------|
| InstCode     | String/text | 20  |   | M | For Kenya use 54, Uganda 56 , Tanzania 55, Rwanda 50 and South Sudan 11         |
| CurrencyCode | String/text | 3   | N | M | The ISO currency code of the transaction Amount                                 |
| CountryCode  | String/text | 2   | N | M | Destination country code as defined by ISO 3166                                 |
| BicCode      | String/text | 50  | Y | O | Destination Bank Swift Bic code. For transactions to be remitted to other banks |
| MobileNumber | String/text | 12  | N | O | In case funds are remitted to a Telkom Wallet recipient Mobile number           |
| WalletName   | String/text | 50  | Y | O | An identifier to identify Telkom Wallet                                         |
| BankCode     | String/text | 5   |   | O | Not Used now                                                                    |
| BranchCode   | String/text | 5   |   | O | Not Used Now                                                                    |
| SenderName   | String/text | 100 | Y | O | A sender name is required for SMS Notification                                  |
| TransAmount  | float       | -   |   | M | Transaction Amount                                                              |



### Response Field Mapping

| Source ServiceTag | Target Field | Data Type | Length | Description<br>Default/Expected Values                                                                               |
|-------------------|--------------|-----------|--------|----------------------------------------------------------------------------------------------------------------------|
| StatusMessage     | Status.Text  | String    | 100    | Success or Failure Message<br>Success → OK {TranId}<br>Error → ERROR {Errorcode} <br>{Optional_ThirdParty_ErrorCode} |

### Request XML

```
<?xml version="1.0" encoding="UTF-8"?>

<S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/">

  <S:Header>

  </S:Header>

  <S:Body>

    <ns0:tranApplyRequest xmlns:ns0="http://ws.equitybank/">

      <accountNumber>100019437XXXX</accountNumber>

      <paymentReference1>Bridge Educational Gaps Project</paymentReference1>

      <paymentReference2>323435445</paymentReference2>

      <paymentReference3>sender@gmail.com </paymentReference3>

      <paymentType>WU</paymentType>

      <remarks>Geoffrey Doe</remarks>

      <agentID>5</agentID>

      <check>AB80w1h2G+SDAL6qwNsVW3HaYEI/
5zvSuPhl+gDLITfzORxsLhK7rP2EBnt6jePpwWWDEahYHsmH9WTZHaNDizhUEK4FLiaH6qrgx06y45zyZvTBKUAHFsox5V04XQOTVO2YQ3JRZYa
QcgRCV+qmDUfQZQmTqPs2rzly5pE1yTRVA/McF/b6d8yip8wQ5C9utq6l58sj2g7WuKeDjIVj4u7C6xsJI+h/
KZXUq348QpYV929wO76pYidJBjLRmJjevdDx7gZZ3NxxnJlB+waBnUBNpsocFyCouqXyYDWWpCk8pYq/vc/
FP1rIKgxLywD9jzFt4galZpM+fc51qw+Lg==</check>

      <fees>0.0</fees>

      <instCode>KE</instCode>

    </ns0:tranApplyRequest>

  </S:Body>

</S:Envelope>
```



```
<tranAmount>72800.0</tranAmount>

<data>518227</data>

<messageID>518227</messageID>

<currencyCode>KES</currencyCode>

<countryCode>KE</countryCode>

<bicCode>EQBLKENA</bicCode>

<bankCode />

<branchCode />

<senderName>Ali, Yusuf</senderName>

</ns0:tranApplyRequest>

</S:Body>

</S:Envelope>
```

>

### Response XML

```
<?xml version="1.0" encoding="UTF-8"?>

<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">

  <soap:Body>

    <ns2:tranApplyRequestResponse xmlns:ns2="http://ws.equitybank/">

      <return>OK|300776091932</return>

    </ns2:tranApplyRequestResponse>

  </soap:Body>

</soap:Envelope>
```



## 1.2.2 Status API

### *Request Field Mapping*

| Source Service Tag | Data Type   | Length | Variable Length | Mand/Opt | Description - Default/Expected Values                                                                                                                                                                    |
|--------------------|-------------|--------|-----------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| messageID          | Text/String | 20     | Y               | M        |                                                                                                                                                                                                          |
| agentID            | Text/String | 10     | Y               | M        |                                                                                                                                                                                                          |
| paymentReference1  | Text/String | 50     | Y               | M        |                                                                                                                                                                                                          |
| check              | Text/String | 200    | Y               | M        | sign and encode:<br>case paymentType<br>optional:<br>salt + MessageID +<br>AgentID +<br>PaymentReference1<br>case paymentType is<br>supplied<br>salt + MessageID +<br>paymentType +<br>PaymentReference1 |
| paymentType        | Text/String | 100    | Y               | O        |                                                                                                                                                                                                          |



### Response Field Mapping

| Source Service Tag | Data Type   | Length | Variable Length | Mand/Opt | Description - Default/Expected Values |
|--------------------|-------------|--------|-----------------|----------|---------------------------------------|
| return             | Text/String | 100    | Y               | M        |                                       |

### Request XML

```
<?xml version="1.0" encoding="UTF-8"?>
<S:Envelope xmlns:S="http://schemas.xmlsoap.org/soap/envelope/">
  <S:Header>
  </S:Header>
  <S:Body>
    <ns0:tranApplyRequestStatus xmlns:ns0="http://ws.equitybank/">
      <messageID>518227</messageID>
      <agentID>5</agentID>
      <paymentReference1>Bridge Educational Gaps Project</paymentReference1>
      <check>AB80uJ1h2G+SDAL6qwNsVW3HaYEI/
5zvSuPhI+gDLITfzORxsLhK7rP2EBnt6jePpwWWDEahYHsmH9WTZHaNDizhUEK4FLiaH6qrgx06y45zyZvTBKUAHFsox5V04XQOT
VO2YQ3JRZYaQcgRCV+qmDUfQZQmTqPs2rzly5pE1yTRVA/McF/b6d8yip8wQ5C9utq6l58sj2g7WuKeDjlVj4u7C6xsJI+h/
KZXUq348QpYV929wO76pYidJLrmJjevdDx7gZ23NxxnJlB+waBnUBNpsocFyCouqXyYDWWPcK8pYq/vc/
FP1rlKgxLywD9jzFt4galZpM+fc51qw+Lg==</check>
    </ns0:tranApplyRequestStatus>
  </S:Body>
</S:Envelope>
```

### Response XML

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/">
  <soap:Body>
    <ns2:tranApplyRequestResponse xmlns:ns2="http://ws.equitybank/">
      <return>OK|300776091932|PAID SUCCESSFUL|MSGID=256454</return>
    </ns2:tranApplyRequestResponse>
  </soap:Body>
</soap:Envelope>
```

## 1.3 Response/Error codes

### 1.3.1 Status error code and description

|  | ERROR_CODE | ERROR_MESSAGE                                          |
|--|------------|--------------------------------------------------------|
|  | 906        | Cut over (end of day) in progress                      |
|  | 911        | Host timed out                                         |
|  | 116        | Insufficient Funds                                     |
|  | 114        | Invalid Account Number                                 |
|  | 902        | Invalid function code                                  |
|  | 115        | Invalid processing code                                |
|  | 904        | Message incorrectly formatted                          |
|  | 908        | Processing host cannot be reached                      |
|  | 907        | Processing host is down                                |
|  | 909        | System Malfunction                                     |
|  | 000        | Transaction Approved                                   |
|  | 119        | Transaction Not Permitted                              |
|  | 180        | Transfer limit exceeded                                |
|  | 121        | Withdrawal limit exceeded                              |
|  | 77         | Payment Transaction Not found                          |
|  | -1         | Validation: System error                               |
|  | -2         | Validation: failed Missing Parameters                  |
|  | -3         | Transaction Failed- Core banking system general error  |
|  | -4         | Transaction Failed- Failed to reach third party system |
|  | -7         | Validation:Transaction Duplicate                       |
|  | -9         | Invalid or Unrecognized Wallet code                    |
|  | -10        | Validation Deposit Limits exceeded                     |
|  | -99        | Validation: Transaction Integrity Failed               |

### 1.3.2 Additional Wallet Error Codes on Status API

| WALLET NAME | ERROR CODE | ERROR DESCRIPTION                                    |
|-------------|------------|------------------------------------------------------|
| Mpesa       | 2          | Limit rule: less than the minimum transaction amount |
| Mpesa       | 7          | Receiver mobile number is in invalid                 |
| Mpesa       | 14         | Receiver mobile number is in an invalid state        |
| Mpesa       | 8          | Would exceed the maximum balance                     |
| Mpesa       | 2040       | Receiver mobile number is unregistered in MPesa      |
| Mpesa       | 5          | M-pesa Validation Failed                             |

## 1.4 Code Samples

### Java

```
package equitybank.ws;

import java.io.File;
import java.net.MalformedURLException;
import java.net.URL;
import javax.xml.namespace.QName;
import javax.jws.WebMethod;
import javax.jws.WebParam;
import javax.jws.WebResult;
import javax.jws.WebService;
import javax.xml.bind.annotation.XmlSeeAlso;
import javax.xml.ws.RequestWrapper;
import javax.xml.ws.ResponseWrapper;

public final class TransApplyClient {

    private static final QName SERVICE_NAME = new QName("http://ws.equitybank/", "TransApply");

    private TransApply_TransApplyImpPort_Client() {

    }

    public static void main(String args[]) throws java.lang.Exception {
        URL wsdlURL = TransApply_Service.WSDL_LOCATION;
        if (args.length > 0 && args[0] != null && !"".equals(args[0])) {
            File wsdlFile = new File(args[0]);
            try {
                if (wsdlFile.exists()) {
                    wsdlURL = wsdlFile.toURI().toURL();
                } else {
                    wsdlURL = new URL(args[0]);
                }
            } catch (MalformedURLException e) {
                e.printStackTrace();
            }
        }

        TransApply_Service ss = new TransApply_Service(wsdlURL, SERVICE_NAME);
```





```
TransApply port = ss.getTransApplyImpPort();

{
    System.out.println("Invoking tranApplyRequest...");
    java.lang.String _tranApplyRequest_accountNumber = "";
    java.lang.String _tranApplyRequest_paymentReference1 = "";
    java.lang.String _tranApplyRequest_paymentReference2 = "";
    java.lang.String _tranApplyRequest_paymentReference3 = "";
    java.lang.String _tranApplyRequest_paymentType = "";
    java.lang.String _tranApplyRequest_remarks = "";
    java.lang.String _tranApplyRequest_agentID = "";

    java.math.BigDecimal _tranApplyRequest_fees = new java.math.BigDecimal("0");
    java.lang.String _tranApplyRequest_instCode = "";
    java.math.BigDecimal _tranApplyRequest_tranAmount = new java.math.BigDecimal("0");
    java.lang.String paramdata = "";
    java.lang.String parammessageID = "";
    java.lang.String paramcurrencyCode = "";
    java.lang.String paramcountryCode = "";
    java.lang.String parambicCode = "";
    java.lang.String parammobileNumber = "";
    java.lang.String paramwalletName = "";
    java.lang.String parambankCode = "";
    java.lang.String parambranchCode = "";
    java.lang.String paramsenderName = "";
    java.lang.String paramcallBackUrl = "";
    String theXsumFields = "";
    if ((paramBicCode != null) && (!paramBicCode.isEmpty())) {
        theXsumFields = vfxSalt + paramAccountNumber + paramBicCode +
paramPaymentReference1
        + paramPaymentType + paramAgentID + paramTranAmount.setScale(2,
RoundingMode.CEILING) + paramCurrencyCode + paramMessageID;
    } else if ((paramWalletName != null) && (!paramWalletName.isEmpty())) {
        paramWalletName = paramWalletName.toUpperCase();
        theXsumFields = vfxSalt + paramWalletName + paramMobileNumber +
paramPaymentReference1
        + paramPaymentType + paramAgentID + paramTranAmount.setScale(2,
RoundingMode.CEILING) + paramCurrencyCode + paramMessageID;
    }
    Base64 base64 = new Base64();
    java.lang.String paramcheck = base64.encode(signData(theXsumFields.getBytes()));
    java.lang.String param_return = port.tranApplyRequest(paramaccountNumber,
parampaymentReference1, parampaymentReference2, parampaymentReference3, parampaymentType,
```



```
paramremarks, paramagentID, paramcheck, paramfees, paraminstCode, paramtranAmount, paramdata,
parammessageID, paramcurrencyCode, paramcountryCode, parambicCode, parammobileNumber,
paramwalletName, parambankCode, parambranchCode, paramsenderName, paramcallBackUrl);
    System.out.println("tranApplyRequest.result=" + param_return);
}
System.exit(0);
}

    public static byte[] signData(byte[] data) {
try {
    Security.addProvider(new BouncyCastleProvider());
    KeyStore ks = KeyStore.getInstance(KEYSTORE_INSTANCE);
    ks.load(new FileInputStream(KEYSTORE_FILE), KEYSTORE_PWD.toCharArray());
    Key key = ks.getKey(KEYSTORE_ALIAS, KEYSTORE_PWD.toCharArray());
    //Sign
    PrivateKey privKey = (PrivateKey) key;
    Signature signature = Signature.getInstance("SHA256withRSA", "BC");
    signature.initSign(privKey);
    signature.update(data);
    return (signature.sign());
} catch (KeyStoreException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
} catch (FileNotFoundException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
} catch (IOException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
} catch (NoSuchAlgorithmException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
} catch (CertificateException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
} catch (UnrecoverableKeyException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
} catch (NoSuchProviderException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
} catch (InvalidKeyException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
} catch (SignatureException ex) {
    Logger.getLogger(TransApplyClient.class.getName()).log(Level.SEVERE, null, ex);
}
    return null;
}
}
```